SINA POURBAHARI

Materials and Metallurgical Engineer | R&D, Metal Recycling and Project Leadership

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PROFILE

Materials and metallurgical engineer (**M.Sc. Politecnico di Milano**) with hands-on experience in **R&D**, **metal recycling**, **casting** and **materials testing**. Currently **leading an R&D project** and an **international team**. **Available from February 2027** and **to travel across Europe**.

PROFESSIONAL EXPERIENCE

Innovation Project Lead, Elihome LLC

Milan, Italy | Aug 2026 to Present (project ends January 2027)

- Coating project stuck for **3 months**: found the **root cause**, project moved to the next stage.
- Mapped **16 technical challenges** in a new **food-contact coating**, with a fix for each, for the **CEO**.
- Own an **AI research project end to end**: brief, task allocation, deadlines and reporting.
- Built an **international team of 5** after **~20 interviews** across **8 countries**.

Graduate Researcher, FluoriTech Lab

Politecnico di Milano | Milan, Italy | Mar 2025 to Aug 2026

- Developed an **acid-free battery recycling route**: **~80% Li** and **~60% Co** at room temperature.
- Estimated **up to 40% lower CO2** than pyrometallurgy; validated on **real industrial scrap**.
- Led **6 campaigns (~30 trials)** and a **9-point time series**, a roadmap for **scale-up**.
- Commissioned a **vacuum and gas line**; completed **3 pump repair cycles**.

Research Assistant, Solidification and Casting Lab

University of Tehran | Tehran, Iran | May 2020 to Sep 2023

- Ran **200+ tensile, hardness and wear tests** on **Mg alloys** and **stir cast composites**.
- **Restored an idle wear tester**: recalibrated it and wrote procedures for **repeatable results**.
- Traced defects to **casting** and **heat treatment** settings (**root cause analysis**).
- Coordinated lab workflows for a **team of 5**; **3D printed fixtures** cut tool costs by **~80%**.

Other Experience

- **Intern, Azerakhsh Metallurgy Service Center** (2022): **heat treatment** effects on copper hardness and conductivity.
- **Teaching Assistant, University of Tehran** (2021): taught **AutoCAD** and **SolidWorks** to **80 students**.

EDUCATION

M.Sc. Materials Engineering and Nanotechnology

Politecnico di Milano, Italy | Taught in English

Thesis: **Mechanochemical recovery of valuable metals from spent Li-ion batteries**

B.Sc. Materials and Metallurgical Engineering

University of Tehran, Iran | **Top 15 in program** (2023) | **Full scholarship**

Thesis: **SiC addition on structure and wear of ZW31 Mg alloy**

SKILLS

Characterization: **SEM**, EDS, **XRD**, XRF, **ICP-OES**, NMR, UV-Vis, **metallography**, tensile, hardness, wear

Process: **casting**, **heat treatment**, **recycling**, **ball milling**, leaching, **root cause analysis**, **DOE**

Project: **project planning**, **team leadership**, **reporting to management**, hiring, documentation

Software: **Excel**, **Microsoft Office** (PowerPoint advanced), **SolidWorks**, **AutoCAD**, MATLAB (certified)

Languages: **English C1** (TOEFL iBT 97), **Italian A2** (currently attending **B1** classes), Persian (native)

Awards: Top 1.5% in the Iranian National University Entrance Exam (144,000 participants)